

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which gene is mutated in Familial Adenomatous Polyposis, and what happens to colorectal cancer risk if it is left untreated?
2. Name two GI anomalies that occur in Down syndrome.
3. What is the key genetic difference between FAP and Lynch syndrome, in terms of how many polyps each one causes?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

Five ways genes go wrong in the GI tract:

Single-gene mutation → inherited polyposis syndromes (FAP, Lynch, Peutz-Jeghers, Juvenile polyposis)

Chromosomal abnormality → Down syndrome, Turner syndrome

Multiple genes + environment → IBD, Celiac disease

Mitochondrial DNA mutation → MNGIE

Somatic / epigenetic change (not inherited) → sporadic GI cancers (KRAS, TP53, methylation)

1 Polypoid syndromes — the exam favourite

These four get confused constantly. Learn the ONE discriminating clue for each, not just the gene name.

Syndrome	Gene	Polyp type & site	Giveaway clue	Cancer risk
FAP	APC (tumour suppressor)	Hundreds-thousands of adenomatous polyps, colon & rectum; also duodenal polyps	Onset adolescence/early adulthood; desmoid tumours	~ 100% CRC if untreated
Lynch / HNPCC	MLH1, MSH2, MSH6, PMS2 (mismatch repair)	Few or no polyps — not a polyposis syndrome	Microsatellite instability ; endometrial/gastric/ovarian cancer also occur	3–5% of all CRC; up to 80% lifetime CRC risk
Peutz-Jeghers	STK11 / LKB1	Hamartomatous polyps, mainly small intestine	Mucocutaneous pigmentation — dark spots on lips/mouth	↑ GI cancers + breast/ovarian cancer
Juvenile polyposis	BMPRI1, SMAD4 (TGF-β pathway)	Hamartomatous polyps, primarily colon	Presents in childhood/ juveniles	↑ CRC, sometimes gastric cancer

2 Chromosomal abnormalities

Condition	Mechanism	GI manifestations
Down syndrome (Trisomy 21)	Extra copy of chromosome 21	Hirschsprung disease (aganglionic segment → obstruction/constipation), duodenal atresia , imperforate anus , esophageal atresia
Turner syndrome (45,X)	Partial/complete loss of one X chromosome	Less common; pyloric stenosis , inflammatory bowel disease

3 Complex polygenic (multifactorial) disorders

Multiple genes, each a small risk, combined with environmental triggers.

	Crohn's disease	Ulcerative colitis
Gene	NOD2 (associated with both)	
Distribution	Any part of the GI tract, patchy	Colon & rectum only , continuous
Findings	Strictures, fistulas	Continuous inflammation, ulcers
Celiac disease		
Genes	HLA-DQ2, HLA-DQ8	

Mechanism	Gluten ingestion → immune-mediated destruction of small intestinal villi in genetically susceptible individuals
Clinical	Malabsorption, diarrhea, anemia; long-term untreated → ↑ risk of small bowel lymphoma

4 Mitochondrial disorder

Gene	TYMP (thymidine phosphorylase)
Mechanism	Enzyme deficiency → toxic nucleoside build-up → mitochondrial damage across tissues
Presentation	Severe GI dysmotility, pseudo-obstruction, malabsorption — plus neurological deficits
Name	Mitochondrial Neurogastrointestinal Encephalomyopathy (MNGIE)

5 Non-inherited changes — cancer drivers

Not passed down; acquired after conception. These drive **sporadic**, not familial, GI cancer.

Type	Mechanism	Example genes / effect
Epigenetic	Abnormal DNA methylation / histone modification	Silences tumour suppressors or activates oncogenes → colorectal & other GI cancers
Somatic mutation	Mutation arising in individual cells after conception	KRAS, TP53 — classic drivers of sporadic colon cancer

P Must-remember pearls

- FAP = **APC** gene, thousands of adenomatous polyps, ~100% CRC risk if untreated.
- Lynch/HNPCC = mismatch repair gene mutation, few/no polyps but up to 80% lifetime CRC risk plus endometrial/gastric/ovarian cancer.
- Peutz-Jeghers = **STK11**, hamartomatous polyps + mucocutaneous pigmentation of lips/mouth.
- Down syndrome GI triad to know: Hirschsprung disease, duodenal atresia, esophageal atresia (+ imperforate anus).
- Somatic mutations (KRAS, TP53) are **not inherited** — they drive sporadic, not familial, GI cancers.

Q Test yourself

1. A 22-year-old has hundreds of colonic adenomas on colonoscopy plus several duodenal polyps. Which gene is most likely mutated?

- A) MLH1
B) APC
C) STK11
D) SMAD4

2. A 45-year-old with only a handful of colonic adenomas develops colorectal cancer; her mother had endometrial cancer. Which mechanism best explains this pattern?

- A) Chromosomal nondisjunction
B) Microsatellite instability from an MMR gene mutation
C) TGF-β pathway disruption
D) A somatic KRAS mutation

3. A teenager with dark pigmented macules around the mouth is found to have hamartomatous polyps in the small intestine. Which gene is implicated?

- A) APC
B) NOD2
C) STK11
D) BMPRI1A

4. A newborn with Down syndrome presents with bilious vomiting and a "double bubble" sign on abdominal X-ray. What is the diagnosis?

- A) Hirschsprung disease
B) Duodenal atresia
C) Imperforate anus
D) Pyloric stenosis

5. Colon biopsy shows transmural, patchy inflammation with skip lesions and a fistula tract. Which IBD subtype is this?

- A) Ulcerative colitis
B) Crohn's disease
C) Celiac disease
D) Microscopic colitis

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1** — B) APC — the classic FAP presentation: hundreds+ adenomas plus extracolonic (duodenal) polyps.
- 2** — B) Microsatellite instability — Lynch syndrome, from an MMR gene mutation; explains the low polyp count but high, multi-organ cancer risk.
- 3** — C) STK11/LKB1 — Peutz-Jeghers syndrome; mucocutaneous pigmentation is the characteristic clue.
- 4** — B) Duodenal atresia — a classic Down syndrome GI anomaly; the "double bubble" is its radiographic sign.
- 5** — B) Crohn's disease — patchy, transmural inflammation with fistulas; ulcerative colitis is continuous and mucosal, confined to colon/rectum.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Genetics GIT Block · MEDucation by Mattin Majid · Source: GIT Genetics LG 1, Dr. Evan Latef Khlef
Study alongside the lecture, not instead of it.